Transfinite Dimension Theory

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Brief History and Motivation

For a regular space X, $\mathrm{ind}X$ is defined recursively as follows:

- $\mathrm{ind}X = -1, \quad X = \emptyset;$
- $\mathrm{ind}X \leq n, \quad (\forall x \in X)(\forall U \in \mathcal{N}_x)(\exists V \in \mathcal{N}_x)$

  $[V \subseteq U \ \ \& \ \ \mathrm{ind}\partial V < n];$
- $\mathrm{ind}X = n, \quad \mathrm{ind}X \leq n \ \ \& \ \ (\forall m < n)[\mathrm{ind}X \leq m \text{ fails}].$

Fact. $\mathrm{ind}\mathbb{R}^n = n$ for $n \in \mathbb{N}$.

Definition

For a separable metric space X, we define dX recursively as follows:

- $dX = -1, \quad X = \emptyset;$
- $dX = 0, \quad X$ has a basis of clopen sets;
- $dX \leq \alpha, \quad X = Z \cup \bigcup\{F_n : n < \omega\}$ where

  $dZ \leq 0,$ each F_n closed and $dF_n < \alpha;$
- $dX = \alpha, \quad dX \leq \alpha$ and $\forall \beta < \alpha,\ dX \leq \beta$ fails.

Results

Theorem

d is a topological invariant.

Theorem

For $n < \omega$, $dX = n$ if and only if $\mathrm{ind}X = n$.

Theorem

If dX exists, dX is a countable ordinal.

Theorem (Subspace Theorem)

If $Y \subseteq X$ and dX exists, then dY exists and $dY \leq dX$.

Theorem

If $dX = \alpha$, then, for any $\beta \leq \alpha$, there is a closed subset $F \subseteq X$ so that $dF = \beta$.

Theorem (Countable Sum Theorem)

If $X = \bigcup\{F_n : n < \omega\}$ where each F_n is closed, then $dX = \sup\{dF_n : n < \omega\}$.

For an ordinal α, we let
$\lambda(\alpha) = \sup\{\beta : \beta \leq \alpha \text{ and } \beta \text{ is a limit ordinal}\}$ and $n(\alpha) < \omega$ the unique natural so that $\alpha = \lambda(\alpha) + n(\alpha)$. Now, define $\Sigma : \mathrm{ON}^2 \to \mathrm{ON}$ by the rule

$$\Sigma(\alpha, \beta) = \max\{\lambda(\alpha), \lambda(\beta)\} + \min\{\lambda(\alpha), \lambda(\beta)\} + n(\alpha) + n(\beta).$$

Theorem (Product Theorem)

Suppose X and Y are so that dX and dY exist. Then $d(X \times Y) \leq \Sigma(dX, dY)$.

Theorem (Addition Theorem)

Suppose X and Y are so that dX and dY exist. Then $d(X \cup Y) \leq \Sigma(dX, dY) + 1$.

For a dimension function e and a limit ordinal α, we say that X is **strongly α-dimensional** if $X = \bigcup\{F_n : n < \omega\}$ where each F_n is closed and $eF_n < \alpha$.

Theorem

Let λ be a limit ordinal, $n \in \mathbb{N}$ and suppose dX exists. Then $dX \leq \lambda + n$ if and only if $X = X_\lambda \cup X_n$ where X_λ is strongly λ-dimensional and $dX_n \leq n$.

Axiomatic Characterizations

If e is a finite dimension function on the class of separable metrizable spaces that satisfies the following, then $e = \text{ind}$.

N.I. $e\{\emptyset\} = 0$;

N.II. $Y \subseteq X$ implies that $eY \leq eX$;

N.III. if $X = \bigcup\{F_n : n < \omega\}$ where each F_n is closed, $eX = \sup\{eF_n : n < \omega\}$;

N.IV. $e(X \cup Y) \leq eX + eY + 1$;

N.V. if $eX < \omega$, then there is a compactification $\hat{X}$ of X so that $e\hat{X} = eX$;

N.VI. if $X \neq \emptyset$ and $eX < \omega$, then there is a base $\mathcal{B}$ for X so that, for every $U \in \mathcal{B}$, $e\partial U \leq eX - 1$.

I. $e\{\emptyset\} = 0$;

II. $Y \subseteq X$ implies that $eY \leq eX$;

III. if $X = \bigcup\{F_n : n < \omega\}$ where each F_n is closed, $eX = \sup\{eF_n : n < \omega\}$;

IV. $e(X \cup Y) \leq \Sigma(eX, eY) + 1$;

V. if eX exists, then there is a compactification $\hat{X}$ of X so that $e\hat{X} = eX$;

VI. if $X \neq \emptyset$ and eX exists, then there is a base $\mathcal{B}$ for X so that, for every $U \in \mathcal{B}$, $e\partial U \leq eX - 1$ when eX is a successor or ∂U is strongly eX-dimensional when eX is a limit.

Questions

In [2], a metric continuum is constructed which has d dimension $\omega + 1$.

Question. Does there exist a separable metric space X with $dX \geq \omega + 2$?

Question. Does (V) hold? Do (I - VI) characterize the d dimension?

References

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[2] M. Urbański.
A new transfinite dimension for metrizable spaces.
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